Principle of Duality

If a compound statement S_1 contains only $\neg$, $\wedge$, and $\vee$ and statement S_2 arises from S_1 by replacing $\wedge$ by $\vee$, and, $\vee$ by $\wedge$ then S_1 is a tautology if and only if S_2 is a contradiction.

For example

(i) The dual of $(p \vee q) \wedge (r \wedge s) \vee \mathbb{F}$ is $(p \wedge q) \vee (r \vee s) \wedge \mathbb{T}$.

(ii) The dual of $p \wedge [\neg q \vee (p \wedge q) \vee \neg r]$ is $p \vee [\neg q \wedge (p \vee q) \wedge \neg r]$.

12.3.5 Logical Equivalence

Definition 12.20

Any two compound statements A and B are said to be **logically equivalent** or simply **equivalent** if the columns corresponding to A and B in the truth table have **identical truth values**. The logical equivalence of the statements A and B is denoted by $A \equiv B$ or $A \Leftrightarrow B$.

From the definition, it is clear that, if A and B are logically equivalent, then $A \Leftrightarrow B$ must be a **tautology**.

Some Laws of Equivalence

1. Idempotent Laws

(i) $p \vee p \equiv p$ (ii) $p \wedge p \equiv p$.

Proof

p	p	$p \vee p$	$p \wedge p$
T	T	T	T
F	F	F	F

Table 12.14

In the above truth table for both p, $p \vee p$ and $p \wedge p$ have the same truth values. Hence $p \vee p \equiv p$ and $p \wedge p \equiv p$.

2. Commutative Laws

(i) $p \vee q \equiv q \vee p$ (ii) $p \wedge q \equiv q \wedge p$.

Proof (i)

p	q	$p \vee q$	$q \vee p$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

Table 12.15

The columns corresponding to $p \vee q$ and $q \vee p$ are identical. Hence $p \vee q \equiv q \vee p$.

Similarly (ii) $p \wedge q \equiv q \wedge p$ can be proved.

3. Associative Laws

(i) $p \vee (q \vee r) \equiv (p \vee q) \vee r$ (ii) $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$.

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